

INVESTIGATION REPORT

Issue 1007: Single-step with a stalled decode stage

openhwgroup/cv32e40p

|                |                                                         |
|----------------|---------------------------------------------------------|
| Document       | RTL-TRIAGE-CV32E40P-1007                                |
| Disposition    | Dead code, proven                                       |
| Record date    | 2026-09-11                                              |
| Classification | Internal working document. Not a verification sign-off. |

1 Status

|               |                                                   |
|---------------|---------------------------------------------------|
| Issue         | openhwgroup/cv32e40p#1007                         |
| Issue type    | Coverage hole referred for disposition            |
| Affects       | cv32e40p                                          |
| Configuration | COREV_PULP=1, FPU=0 (RTL master 6033d2b)          |
| Disposition   | Dead code, proven                                 |
| Severity      | Not assessed                                      |
| Agreement     | Same disposition as the independent human record. |
| Evidence      | 3 cited, 2 present, 1 not retained                |

Disposition meaning. The code cannot be reached in any state permitted by the stated model and assumptions, established by an unbounded proof.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

cv32e40p\_controller.sv:675 is the 'if (id\_ready\_i)' inside the DECODE single-step block (debug\_single\_step\_i && !debug\_mode\_q, after the debug-entry and interrupt branches were not taken). The uncovered arm is id\_ready\_i==0. Proven unreachable with an unbounded abc pdr proof (124.5 s) of the exact guard: whenever the controller is in DECODE with a valid instruction, single-step set and debug mode clear, id\_ready\_i is 1. Mechanism: dcsr.step can only be set while in debug mode, and the core leaves debug mode through dret -> FLUSH\_EX -> FLUSH\_WB -> DECODE with an empty EX/WB pipeline; the single-stepped instruction is therefore the first instruction after the flush, so no load-use, jalr, misaligned or CSR/APU hazard from a previous instruction can stall ID (id\_ready = ~stalls & ex\_ready, all sourced from older instructions), and the same cycle sends the FSM to DBG\_FLUSH so no second instruction ever decodes under single-step. Anti-vacuity: the single-step block itself and its id\_ready\_i==1 arm are reached by the cover engine in 16 cycles, and DECODE with id\_ready\_i==0 is reached in 7 cycles (so the stall path exists, just not under single-step). The hole cover for id\_ready\_i==0 under single-step stayed unreached to depth 40 (cover job then timed out on that last cover, consistent with the proof).

3 Conditions

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

## 5 Action

### Waiver text, to be applied to the coverage exclusion

cv32e40p\_controller.sv:675 'if (id\_ready\_i)' under 'debug\_single\_step\_i & ~debug\_mode\_q' in DECODE: the AllFalse arm (id\_ready\_i==0) is formally unreachable (abc pdr unbounded proof, runs/v1-cv32e40p-1007/formal/p\_line675). Single-step is only active for the first instruction after leaving debug mode, when EX/WB are empty, so no stall source exists; the same cycle moves the FSM to DBG\_FLUSH. The enclosing block is reachable (cover step 16), so the proof is not vacuous. Exclude the id\_ready\_i==0 arm from branch coverage (open-source proof replaces the Cadence SLEC equivalence argument).

### Checker used

checker\_1007.sv

## 6 Evidence

| Ref | Type           | Path, relative to the run directory  | File     | Result recorded                                                                                                                                                                              |
|-----|----------------|--------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E1  | proof          | formal/p_line675/sby.log             | present  | abc pdr PASS unbounded (124.5 s): DECODE && instr_valid && !dbg-entry && !irq && debug_single_step_i && !debug_mode_q -> id_ready_i                                                          |
| E2  | proof          | formal/cover/sby.log                 | present  | cover: single-step block reached at step 16, its id_ready arm at step 16, DECODE&&!id_ready at step 7 (3 of 4 covers); the hole cover unreached to depth 40 (job TIMEOUT on that cover only) |
| E3  | counterexample | formal/cover/job/engine_0/trace1.vcd | not kept | witness of the single-step block being entered (rendered in cover_single_step_block_reached.png)                                                                                             |

*Paths are relative to runs/v1-cv32e40p-1007. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.*

### What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.

*Artefact(s) E3 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.*

## 7 Limitations and scope

- The proof establishes unreachable under the model and the environment assumptions named in Status and Conditions. It does not establish that the surrounding logic is correct.
- This is an agent investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

## 8 Provenance and measurement

**Run directory** runs/v1-cv32e40p-1007

**Title source** Title and configuration read back from reports/issues/cv32e40p\_issue\_1007.pdf, the previously published report for this issue.

**Inputs shown** title, body(links stripped), labels

|                  |                                                                                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Human record     | Dolphin Design 2024 (issue body): waived in v2 after Cadence JasperGold SLEC showed removing lines 675/703 leaves outputs unchanged |
| Agent confidence | 0.9. This is the investigating agent's own estimate. It is not a measurement and not a reliability figure.                          |
| Iterations       | 1                                                                                                                                   |
| Run by           | Claude directly (no headless agent)                                                                                                 |

Measured tool and step time

| Step                                       | Measured          |
|--------------------------------------------|-------------------|
| read issue + locate line 675 + signal list | 5.0 min (300 s)   |
| write assert + 4 covers                    | 4.0 min (240 s)   |
| abc pdr proof                              | 2.1 min (125 s)   |
| cover run (to depth 40)                    | 10.3 min (616 s)  |
| report                                     | 5.0 min (300 s)   |
| Total                                      | 26.4 min (1581 s) |

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

Generated by reporting/scripts/issue\_document.py from runs/v1-cv32e40p-1007/verdict.json. The generator typesets recorded data and checks that cited files exist. It does not run, rerun or validate any check, and it cannot supply missing evidence.